

Credit Card Customer & Revenue Analysis Dashboard

Business Intelligence Project using SQL & Power BI

Tools Used

- SQL (PostgreSQL)
- Power BI
- DAX
- Excel (CSV Dataset)

Project Type

- Business Intelligence | Data Analytics | Dashboard Development

Business Problem

Financial institutions generate large volumes of customer and transaction data every week. Without an interactive reporting system, it becomes difficult to monitor business performance and identify trends.

Project Objective

- This project aims to develop an interactive dashboard that helps stakeholders:
- Monitor overall revenue and transaction performance
- Analyze customer demographics and spending behavior
- Identify high-performing credit card categories
- Track customer acquisition and activation metrics
- Support data-driven business decision making

Project Workflow

Collected customer and transaction datasets in CSV format.



Imported the data into SQL database.



Connected SQL database with Power BI.



Created data model and relationships



Created DAX measures and calculated KPIs.



Designed interactive dashboards.



Generated business insights and recommendations.

Project Workflow

- Collected CSV datasets
- Created SQL database and imported datasets
- Connected SQL database to Power BI
- Created data model and relationships
- Developed DAX measures and calculated columns
- Built interactive dashboards
- Generated business insights and recommendations

Dashboard Overview

The project consists of two interactive dashboards that provide a comprehensive view of customer behavior, transaction activity, and business performance.

1.Customer Dashboard

KPIs:

Customer Revenue

Customer Segmentation

Revenue by Gender

Revenue by Income Group

Revenue by Education

Revenue by Card Category

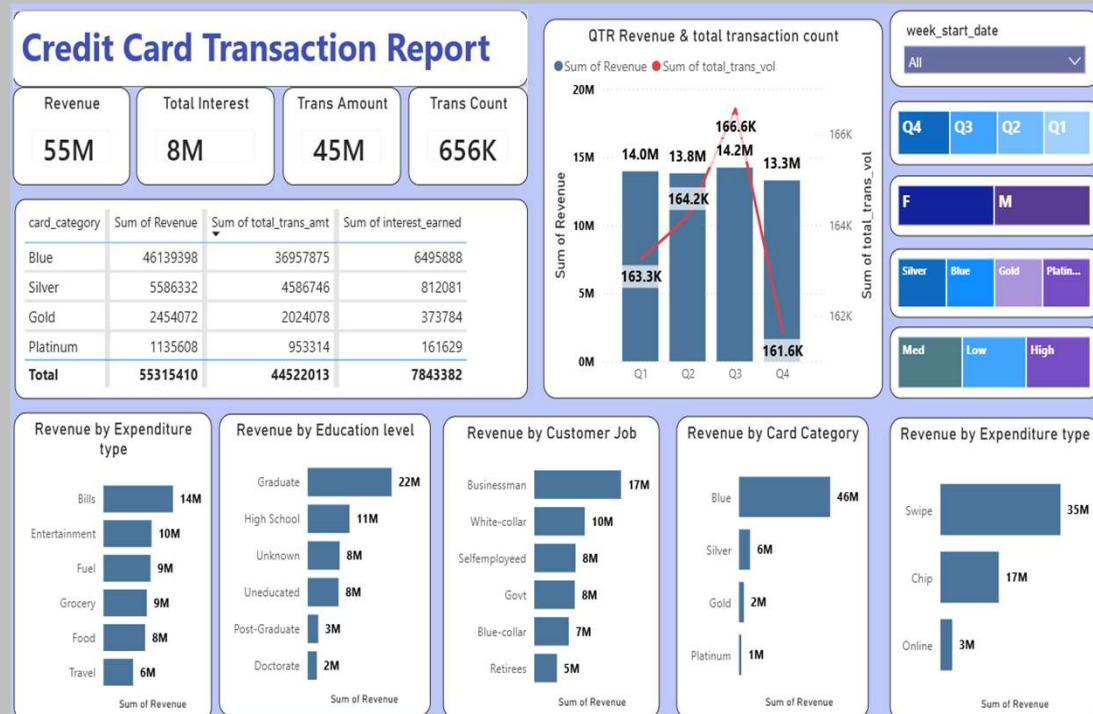
Revenue by State



Dashboard Overview

Transaction Dashboard KPIs

Total Revenue (\$55M)
Transaction Amount (\$45M)
Interest Earned (\$8M)
Transaction Count (656K)
Weekly Revenue Trend
Activation Rate (57.5%)
Delinquency Rate (6.06%)



BUSINESS INSIGHTS

● The business generated approximately **\$55M in total revenue**, with **\$45M** coming from transaction activity and **\$8M** from interest earned.

Business Impact:

Transaction-based spending remains the primary revenue driver, while interest income contributes a stable secondary revenue stream.

Recommendation:

Increase customer transaction frequency through reward programs and cashback offers to maximize revenue growth.

● **Blue Card** customers contributed approximately **\$46M**, accounting for nearly **84%** of total revenue.

Business Impact:

The standard Blue Card portfolio represents the company's largest and most profitable customer base.

Recommendation:

Identify high-spending Blue Card customers and promote premium card upgrades to increase customer lifetime value.

● Swipe transactions generated approximately **\$35M**, whereas Online transactions contributed only **\$3M**.

Business Impact

Customers still rely heavily on physical card usage, indicating relatively low adoption of digital payment channels.

Recommendation

Encourage digital payment adoption through cashback offers, reward points, and exclusive online merchant partnerships.

● The **40–50 years** age group generated approximately **\$25M**, representing the highest revenue contribution among all age segments.

Business Impact

Middle-aged customers appear to have the highest purchasing power and financial engagement.

Recommendation

Prioritize retention campaigns and premium card offerings for this customer segment.

● Texas, New York, and California collectively contributed approximately **68% of total revenue**.

Business Impact

Revenue is highly concentrated in a few key markets, indicating that business performance depends significantly on these states.

Recommendation

Continue investing in these high-performing markets while identifying growth opportunities in lower-performing states to diversify revenue sources.

● The overall customer activation rate is **57.47%**.

Business Impact

Nearly half of acquired customers have not yet activated their credit cards, representing untapped revenue potential.

Recommendation

Implement onboarding campaigns, activation reminders, and welcome incentives to encourage newly acquired customers to activate and use their cards.

● Customers in the **High Income** category generated approximately **\$29M**, significantly outperforming Medium and Low income groups.

Business Impact

Higher-income customers contribute disproportionately to business revenue.

Recommendation

Introduce exclusive benefits and personalized offers to improve retention of high-income customers.

● Quarter 3 recorded the highest revenue (**\$14.2M**) and transaction volume (**166.6K**), whereas Quarter 4 showed a decline.

Business Impact

Revenue and customer spending exhibit seasonal variation.

Recommendation

Investigate factors contributing to the Q4 decline and introduce promotional campaigns to sustain customer spending.

● The overall delinquency rate is **6.07%**, with self-employed customers showing comparatively higher delinquency than other customer segments.

Business Impact

Higher delinquency within this segment may indicate elevated repayment risk, which can affect portfolio quality and increase potential credit losses.

Recommendation

Strengthen credit risk monitoring for self-employed customers through enhanced credit assessments, proactive repayment reminders, and tailored risk management strategies.

● Weekly revenue remained relatively stable throughout the year, with periodic peaks and declines rather than a continuous upward trend.

Business Impact

Customer spending remained consistent, although short-term fluctuations indicate the influence of seasonal factors and promotional activity.

Recommendation

Monitor weekly performance continuously to identify recurring seasonal patterns and evaluate the effectiveness of marketing campaigns.

● The Customer Satisfaction Score (CSS) is **3.19**, indicating moderate customer satisfaction.

Business Impact

There is an opportunity to improve customer experience and increase loyalty.

Recommendation

Analyze customer feedback and improve reward programs, customer service, and digital experience.